Nawaf Alampara

pvt.nawaf@gmail.com
Homepage | **Github** | **LinkedIn** | **HuggingFace** | **Google Scholar** | **CV as PDF**

Summary

I am second-year PhD student, working with Dr. Kevin Maik Jablonka. I'm building machine learning systems to speed up scientific research. I love projects that involve both research and building the tooling that enables and accelerates that research. Lately, I've been analyzing general-purpose AI models/systems to understand their limitations in scientific applications (where they fail) and interpret them to uncover why they fail. My goal is to use these insights to design AI systems that aren't just impressive on benchmarks but truly impactful for advancing science and research.

Research Interests

- Evaluating and interpreting scientific intelligence in AI models
- Task adaptation of general-purpose models
- AI systems for scientific discovery
- Inverse materials design
- Generative models for molecules and materials

Education

Nov 2023 – Present
PhD Digital Chemistry / ML for Science
Friedrich-Schiller-Universität Jena, Jena, Germany
Advisor: Dr. Kevin Maik Jablonka
Research Focus: Evaluation and development of foundation models for scientific applications, AI agents for discovery workflows.

2018 – 2020
MSc Energy Science
Indian Institute of Technology Bombay, Mumbai, India
GPA: 8.55/10.0
Thesis: **Defects and Dopants in Cu_2O - A DFT study** (Received highest grade)
Advisors: Prof. K R Balasubramaniam, Prof. D Monder

2015 – 2018
BSc Physics
Birla Institute of Technology Mesra, Mesra, India
GPA: 8.5/10.0

Experience

Nov 2023 – Present
PhD Research Activities
Friedrich-Schiller-Universität Jena, Jena, Germany

Jun 2025 – Present
Google Summer of Code 2025 contributor with DeepMind
DeepMind, Remote
+ SciResearchBench: Framework to evaluate the reasoning and scientific research capabilities of Gemini models.

Nov 2023 – Apr 2024
AI Research Contractor (Part-time)
Stability AI, Remote
+ Dataset preparation for domain specific large language foundation model for chemistry. + Benchmarking foundation models on domain specific tasks.

Jun 2021 – Sep 2023
Research Engineer
QpiAI Technologies Pvt. Ltd., Bangalore, India
+ Lead the development of QpiVoltaET: A cloud based platform to accelerate the discovery of materials/molecules using accurate and efficient neural network models as an alternative to first principle simulations. + Implemented end to end pipeline for

object detection in high resolution and low resolution images using bare torch vision and pytorch. Worked with clients to implement end-to-end computer vision pipelines for different use cases. + Responsible for development of end to end Deep Stream containers for real time video analytics. Build new gstreamer plugins(C++ deepstream plugins), monocular depth estimation deep learning models for safety applications.

Selected Publications

1. **A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists**
   Adrian Mirza †, **Nawaf Alampara** †, Sreekanth Kunchapu †, Martiño Ríos-García †, ..., Kevin Maik Jablonka.
   Nature Chemistry, 2025. DOI: 10.1038/s41557-025-01815-x
2. **Probing the limitations of multimodal language models for chemistry and materials research**
   Nawaf Alampara, Mara Schilling-Wilhelmi, Martiño Ríos-García, ..., Kevin Maik Jablonka.
   Nature Computational Science, 2025. DOI: 10.1038/s43588-025-00836-3
3. **Less can be more for predicting properties with large language models**
   Nawaf Alampara, Santiago Miret, Kevin Maik Jablonka.
   2024. (**AI4Mat-Vienna 2024 spotlight**)
4. **ChemPile: A 250GB Diverse and Curated Dataset for Chemical Foundation Models**
   Adrian Mirza, **Nawaf Alampara**, Martiño Ríos-García, ..., Kevin Maik Jablonka.
   2025.
5. **General purpose models for the chemical sciences**
   Nawaf Alampara †, ..., Kevin Maik Jablonka.
   2025. (in preparation)
6. **Lessons from the trenches on evaluating machine-learning systems in materials science**
   Nawaf Alampara †, Mara Schilling-Wilhelmi †, Kevin Maik Jablonka.
   Computational Materials Science, 2025. DOI: 10.1016/j.commatsci.2025.114041
7. **Formation of an extended defect cluster in cuprous oxide**
   G Aggarwal, S Chawla, AJ Singh, **Nawaf Alampara**, D S Monder and K R Balasubramaniam.
   Journal of Physics D: Applied Physics, 2024.

(† Equal contribution)

Patents

1. **System and method for performing accelerated molecular dynamics computer simulations with uncertainty-aware neural network**
   Inventors: **Nawaf Alampara**, Aswanth Krishnan, Nagendra Nagaraja.
2. **Lithium iron phosphate (LFP) cathode active materials and method for deposition of the same**
   Inventors: Manjunath Gangaiah, **Nawaf Alampara**, Aswanth Krishnan.
3. **System and method for sensor position optimization for autonomous vehicles**
   Inventors: Amlan Mukherjee, Arun Sehrawat, Nagendra Nagaraja, Aswanth Krishnan, **Nawaf Alampara**, ...

Relevant other Research Projects

Mar 2021 – Oct 2021
EnergyNet: Geometric deep learning model for Structure-Property Prediction
Advisor: Prof. Mayank Baranwal
Developed a GNN model predicting adsorption energy (**Top 7 in NeurIPS OC20 Competition**).

Jan 2020 – Dec 2020
Ab-initio Thermodynamics of Defects in Cu_2O (MSc Thesis)
Indian Institute of Technology Bombay, Mumbai, India
Advisors: Prof. K R Balasubramaniam, Prof. D Monder
Investigated intrinsic defects, dopants (alkali, H), and defect thermodynamics in Cu_2O using DFT (Quantum ESPRESSO) and Boltzmann transport. Identified formation mechanisms of complex defects.

Jan 2020 – Jun 2020
Tight Binding Models for Nanoscale Systems
Indian Institute of Technology Bombay, Mumbai, India
Advisors: Prof. Ashwin A. Tulapurkar, Prof. Bhaskaran Muralidharan
Calculated band structures of graphene nanoribbons. Modeled SSH chains (Python).

May 2017 – Jul 2017
Fabrication of SiC Schottky diode
Centre for Nanoscience, BDU, Tiruchirappalli, India
Advisor: Prof. K. Jeganathan
Fabricated 4H/6H-SiC Schottky diodes and characterized electrical parameters.

Skills

Programming Languages	Python (primary), Golang, JavaScript
ML Stack	PyTorch, PyTorch Geometric, HuggingFace, PyTorch Lightning, OpenCV, DeepStream
Tech Stack	Git, Docker, FastAPI, Flyte, MLflow
Scientific Packages	Quantum ESPRESSO, VASP, ASE, Pymatgen
Additional	Experience with CI/CD, test-driven development, high-performance computing (HPC), finetuning and quantization of LLMs

Reviewer

- **Workshops (12 papers)**: AI4Mat-Vienna 2024, NeurIPS-AI4Mat 2024, ICLR-AI4Mat 2025, ICANN 2025
- **Journals (1 paper)**: Communications Chemistry

Honors and Awards

- **Inspire Fellowship (SHE)** - Awarded by Dept. of Science and Technology, Govt. of India
- **Summer Research Fellowship (SRF 2016)** - Awarded by Indian Academy of Sciences